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Topic: Inter-cloud Architecture and Service Application Brokering.

INTRODUCTION

Cloud computing has been attracting the interest of the information and communications Technology community since 2007 resulting in a massive amount of industry developments. It has been seen as the next logical step after Grid computing on the way to utility computing, where computing resources are available as subscription utility service just like water and electricity. Cloud computing has been defined as a type of parallel and distributed system providing virtualized resources in a pay-as-you-go fashion over the Internet.

From business perspective, it is widely viewed as an economical model for renting technical resources.

However, the standard cloud computing model, where a client utilises a single cloud data centre, introduces several challenges. A cloud service unavailability can leave thousand of customers relying solely on it without access to essential and paid for resources.

DEFINITIONS

Cloud computing is a novel area of research and still faces certain terminological ambiguity. The use of Inter-Clouds is even newer, and many works in the area use several terms interchangeably. Inter-cloud computing has been formally defined as "A cloud model that, for the purpose of guaranteeing service quality, such as the performance and availability of each service, allows on-demand reassignment of resources and transfer of workload."

The seminal works on enter-clouds also implicitly express similar definitions opportunistic nature of the provisioning within an enter-cloud that allows for achieving QoS and quality of experience targets in a dynamic environment.

ARCHITECTURAL TAXONOMY

enter-cloud computing is an endeavour that implies interconnecting multiple cloud providers infrastructure. The extent to which providers would voluntarily lend their infrastructure within an enter-cloud depends on the political and financial incentives they have to do so.

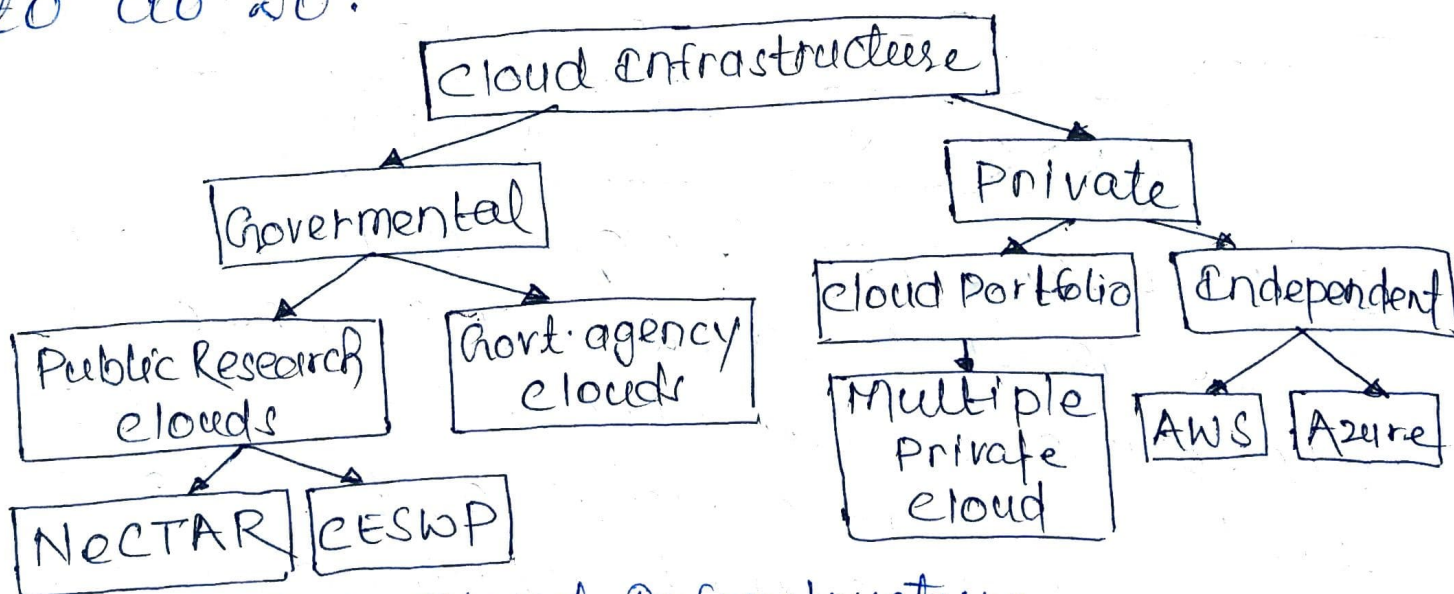


Fig1: Cloud Infrastructure

- Governmental: Owned and utilized by a government or nonprofit institution. Example for this like are science community clouds like Australia's National eResearch Collaboration Tools and Resources (NECTAR) and Canada's cloud-Enabled space Weather Modelling and Data Assimilation Platform (CESWP).
- Private: Owned by a private organisation. Private clouds can be further classified as follows:
 - Cloud Portfolio: When the cloud is a part of a portfolio of clouds belonging to the same organisation. EX- Multiple Clouds.
 - Independent: Separate cloud infrastructure that is not a part of a portfolio of clouds.

TAXONOMY OF INTER-CLOUD APPLICATION BROKERING MECHANISMS

There are several mechanisms for implementing application-specific brokering as depicted. At the first level, we differentiate as to who is responsible to provide implementation of the application broker.

The directly managed brokering mechanisms are mostly used when there is no mediator between the applications and the set of utilized clouds. Directly managed brokers are hosted separately and need to keep track of the performance characteristics of the application themselves. It is the responsibility of the application developers to develop such brokers in a way that meets the availability and dependability requirements.

APPLICATION CENTRIC PERSPECTIVE TO INTER-CLOUDS

Clouds as a deployment environment are general purpose and cover the whole spectrum of distributed application. Cloud providers have always delivered computing resources to enterprises. These many data base centric enterprise applications are deployed within clouds. Being an inherently distributed environment, clouds are also suitable for standard grid and cluster job-based/

Brokers using externally managed mechanisms usually are not hosted in house and are a part of the system entities that facilitate the enter-cloud.

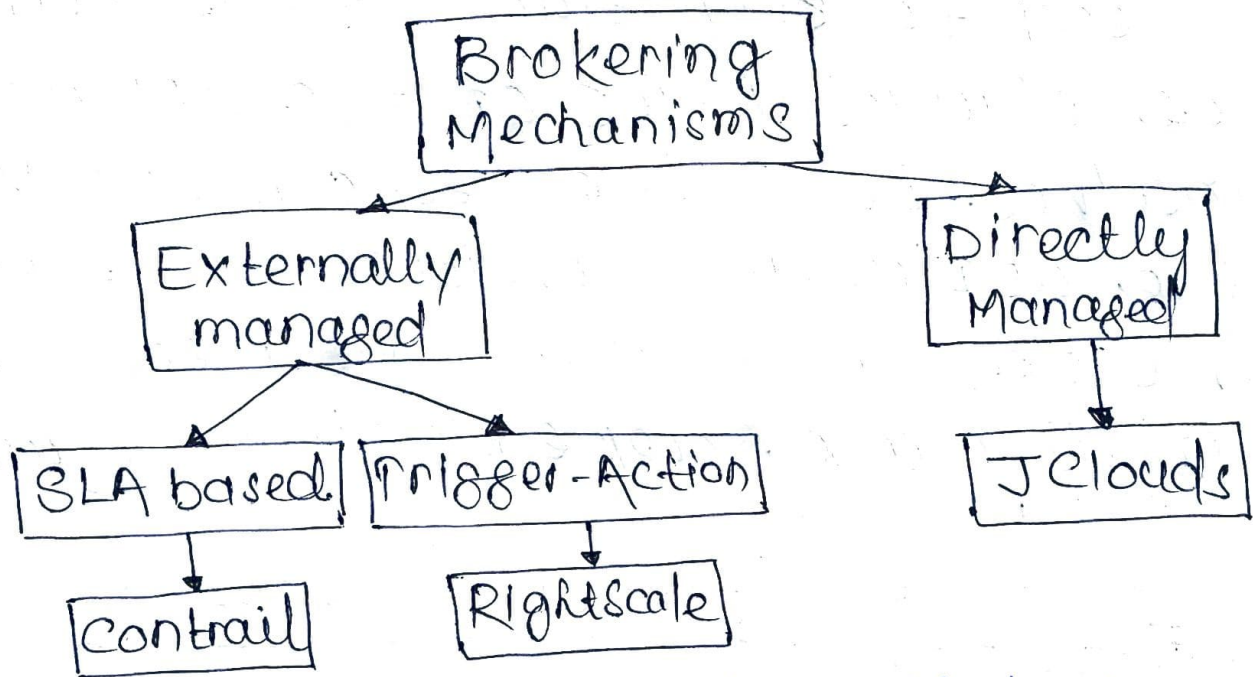


Fig 2: Application Brokering Mechanisms

- SLA Based: SLA ensures cloud providers meet certain enterprise-level requirement and provide customers with a clearly defined set of deliverables.
- Trigger-Action: A trigger becomes active when a predefined condition considering the externally visible application performance indicators becomes true. An action is executed when a correspondent trigger becomes active.

applications. Research community clouds like Australia's NECTAR and Canada's CESWP have made the use of clouds for resource intensive jobs apparent, though there are still concerns about the worse performance of clouds for resource intensive jobs apparent.

REQUIREMENT FOR INTER-CLOUD ENVIRONMENT

Enter-cloud environment is the diversification in terms of vendors and locations. It is however important that this is done without violating the non-functional application requirements. Not all enter-cloud environment allow for such control. Thus in the rest of this section we define several requirements for enter-cloud environment and discuss for which of the main classes of application these are important.

Data Location Awareness: Ideally the persistent data and the processing units of an application should be in the same

data centre and should be connected with a high speed network. If they reside on remote clouds the performance penalty could be grave.

This consideration is especially important for interactive applications and periodical jobs using huge data bases.

Pricing Awareness: A common consideration for all types of applications deployed in multiple clouds is the pricing.

SUMMARY

Through cloud computing itself has many open problems, researchers in the field have already made the leap to envision inter-cloud computing. Their goal is to achieve better overall Quality of Service (QoS), reliability and cost efficiency by utilizing multiple clouds. Inter-cloud research is still in its infancy and the body of knowledge in the area has not been well defined yet.